

Project Definition Document

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1.0 Project Charter (15%)

1.1 Business Need

“Use gamification to incentivize and reward young people for their learning and engagement with healthy money habits. Solutions could incorporate understanding of investment risk” ~ Domine Brockes, Dominic Marcus Financial Consultancy Ltd.

Good personal finance is an important part of one's life as to achieve prosperity, it must be well managed.^[1] However, this knowledge is lacking in young people.^[2] As the younger someone learns these financial skills, the better it is for them,^[3] helping young adults improve their financial literacy will set them up for a better life. As gamification has been shown to be an effective learning tool,^[4] using it would be a fun and engaging way to convey this vital knowledge whilst allowing for high retention rates.^[5] Finally young people are wanting to increase their financial knowledge,^[6] showing that not only is there a problem that needs to be resolved, but that the people that it's affecting want to have the tools needed to fix this. The only missing link is the tool to teach them, hence the problem statement that has been issued.

1.2 Scope

The scope of the statement could potentially cover investment returns or business finance, however there is a key focus on helping young people with understanding their finances related to their first waged job. As such, the core areas that should be focused on in particular are understanding one's payslip, budgeting and creating savings.

Retirement planning should be a strong aspect in the solution. Not only is it one of the typical areas that clients of Dominic Marcus Financial Consultancy want help with,^[7] it's also one of the areas in life that young people should be paying attention to but don't due to a lack of knowledge.^[8] As this has life long impacts to an individual's retirement plans, getting it right early is important. Gamification should be incorporated into the structure and design of the solution to be the primary way that these aspects are taught.

1.3 Constraints

1.3.1 Time

The project must be completed by the end of the Semester. As this is a hard deadline, all other constraints must accommodate this, either by reducing scope or quality to satisfy it. Due to this deadline, any future expansions/developments must occur outside of this to accommodate the increase in scope.

1.3.2 Cost

It is not expected that there will be any costs related to this project's development. The team can opt for paid services such as hosting sites and APIs if they so choose but this should be outlined and justified in section 2.4.2. When considering future development, a new cost analysis needs to be conducted to re-evaluate this constraint.

1.3.3 Scope

The project should incorporate the core learning aspects outlined above and gamification elements. Other than this, the team is able to vary this constraint as needed. Any future features would constitute an increase in scope. As to not decrease quality, the time frame to work must be re-evaluated.

1.3.4 Resources

Six team members should divide work equally amongst themselves to complete the required documentation and project development. All University of Surrey resources are available as well as any found through independent research, but these must be referenced appropriately. More resources may be needed when developing new aspects of the solution outside of the initial minimum viable product. This could include an increase in man-power or more technical resources that could be implemented.

1.3.5 Risk

As a tool to help young people learn healthy financial habits, the main risk associated with this is teaching them the incorrect information. This could lead to end-users getting into financial hardship or legal trouble with HMRC. As such, the utmost care needs to be taken that all information provided by the solution is up-to-date and relevant to the financial laws of the operating country. When adding any new expansion to these constraints, an in depth risk analysis must be conducted to be certain that new unnoticed risks do not negatively impact the development or release.

1.4 Evaluation Criteria

- Includes gamification elements to incentives users to learn
- Teaches users healthy financial habits in the following areas:
 - Pensions/retirement planning
 - Understanding the payslips
 - Budgeting & personal savings
- Solution is aligned with SDG4 (quality education) & SGD8 (decent work and economic growth)

1.5 Potential Stakeholders

Stakeholders that should be considered include: the project Sponsor, Stella Kazamia, and Dominic Brockes from Dominic Marcus Financial Consulting Ltd as these are the individuals overseeing and commissioning the project respectively. The end-users must also be considered throughout the development process to ensure that the end product is an engaging and effective learning tool that they can use optimally.

2.0 Project Description (10%)

2.1 Introduction

Our solution to this problem is a web-based learning platform that teaches users various personal finance skills through learning pathways. The users will learn the information using a lesson and quiz format where each quiz will first teach the user the needed information before quizzing them on it. Gamification will be used as an incentive to keep learning. By completing a quiz, the user will earn coins that they can use to buy items for their companion via the shop, and a leaderboard.

2.2 Stakeholders

Including the stakeholders mentioned in 1.5, more potential stakeholders include other online learning platforms that host similar content, such as Coursera^[1] & FutureLearn,^[2] and the companies that created the APIs that will be used.

2.3 Scope

2.2.1 Current Scope

- An initial tutorial that shows the user how to navigate through the app, and recommends the most relevant learning material.
- Three pathways, each containing three subsections, covering different fundamental financial concepts: Budgeting, Pensions, and Taxes.
- Quizzes to teach and test the user on these topics.
- A selection of tools utilising the APIs to calculate interest rates and similar values.
- A leaderboard displaying monthly and lifetimes points earned per user, as well as a shop to unlock virtual rewards, serving as the gamification aspects.

2.2.2 Future Scope

- The ability to add other users as friends.
- A real-time quiz aspect where multiple users compete for the highest score.
- Compatibility with mobile devices
- Modified Django admin interface to allow administrators to create new quizzes.
- Two further pathways: Banking, and Loans.

2.4 Constraints

Along with the constraints: two APIs must be used, it must run in a docker container and have a real-time client-server streaming component, the following are other identified constraints.

2.4.2 Cost

No costs are anticipated for this project as all technologies are free to use. However, care must be taken that all terms & conditions are adhered to when using external software so that this remains the case throughout the project. See 3.3.2 and 8.1 for more details.

2.4.3 Schedule

The scope of the project has been fully scheduled in 7.0 such that it's highly possible to complete in the 12 week timeframe. Some key deadlines are that the PDD and Project Pitch (PP) are to be completed in week 7, development should begin in week 6 and end in week 11 and the Final Audit Report (FAR) will begin in week 9 to be finished by week 12.

2.4.3 Data Privacy & Legal Constraints

GDPR & Data Protection laws need to be complied with at all times, with user accounts being of the highest concern. This is primarily being addressed through the use of APIs shown in 3.3.1 and discussed more in 4.3.

2.5 Dependencies

The project will depend on the use of the following APIs: Google Identity Services,^[3] and Mortgage Calculator API.^[4] Our project will be built and run using the following technologies: Gitlab, Django, Python, HTML, CSS, Bootstrap, SQLi, and Docker. Figma^[5] and draw.io^[6] have also been used during the design and planning phases. See 3.0 for more details.

2.6 Risks

Projects such as this have many risks that need to be mitigated for a successful completion. This includes: scope creep, as new ideas are brought to light throughout development, difficulties with APIs implementation, and inactive/uncooperative team members. A more in-depth risk analysis can be found in 8.0 along with their mitigations.

2.7 Feasibility

The project will all be implemented in Django, providing the team with the resources needed to create the minimum viable product as discussed above. Two APIs will be used, with the possibility of adding more if time permits, in order to keep development time within our constraints. For similar reasons, our initial product will be kept to three quiz pathways and two calculator tools. In terms of cost, development is possible because there is no cost associated with the project.

Overall, the project is feasible due to the technologies used being easily accessible for development, and the basic contents of the website being kept to a minimum while still representing the full functionality of the app.

3.0 Technical Approach (10%)

3.1 Web Framework - Django

3.1.1 Justification & Limitations

The primary reason Django was selected was due to its high level of familiarity within the team, with everyone having prior experience working with it. This cut the time needed for researching new technologies as well as debugging & coding time during development, increasing the time allotted for quality control.

The secondary reason was its versatility. Django has many features such as the in-built user class and editing the database via models, which while streamlining the development process, will allow for easier changes when implementing future scope.

The lack of customizability with regards to the file structure could be seen as a negative if the team were less experienced using the framework. However this is a non-issue and so won't need to be mitigated.

3.1.2 Database

The database system that will be used in development is SQLite3. It's lightweight and easy to set up, making it perfect to use while developing the application. However, in future, it would be worth considering alternatives like PostgreSQL for better security and the handling of more data.^[1]

A general plan for the structure of our Django project is laid out in Figure 1 (see *in* 9.3).

3.2 Front-end Framework - Bootstrap

Apart from team familiarity, it matches the visual style currently being aimed for, creating a modern-looking webpage without needing a high competency level in languages such as CSS. The framework also includes a JavaScript library for a more interactive site and engaging for the user, something we will need when implementing our quiz system.

Bootstrap isn't perfect; it has a fixed number of options for some components (buttons, alerts etc.) that don't fit our colour scheme. To ensure our frontend has a cohesive look the team will need to override some of the CSS options of some Bootstrap components to suit the design.

3.3 APIs and Streaming Component

3.3.1 Google Identity Services (GIS) API^[2]

This API allows a user to create an account and sign in using an existing Google account, as a 'Sign in with Google' button can be viewed as a more trusted & secure way to sign in,^[3] helping with user signups, with similar APIs being added in future scope. When implementing this API, the team must ensure it works as intended with the built in Django users and the customised extended user model, with the experience being identical no matter how a user's account was created.

3.3.2 API Ninjas Mortgage Calculator API^[4]

The mortgage calculator API by API Ninjas is the second API and it takes the total size of a mortgage, its interest rate (fixed rate) and term length and returns monthly & annual payments, as well as total interest paid. This was included to allow the user to get an idea of the cost of a mortgage and the impact it can have on a budget. A limitation of this API is that it only allows 10,000 free API calls per month, so a limit on how many times a user can make API calls may need to be introduced in the future.

3.3.3 Streaming Component

To meet the streaming component requirement, a real-time leaderboard will be included that updates as users earn points. It will update the totals of each user and their position on the leaderboard as they complete quizzes.

3.4 Prospect for Future Development & Deployment

3.4.1 Deployment

At the end of the module, the project will be submitted in a docker container. This, alongside Django's easy configuration, will make it easy for the project to be deployed in the future.

3.4.2 Future Development

Thanks to the modularity provided by Django's sub-apps, streamlining the introduction of new features, or removing outdated ones, if development were to continue beyond the module. Future improvements could include introducing more finance tools, different quiz styles and more avatar customisation. More account options could also be added such as signing in with other services (e.g. Apple, etc.).

4.0 Evaluation and Success Criteria (10%)

4.1 Project Criteria

The project must include the following features upon completion:

- Web application using Django as a framework such that it can be run inside a docker container
- Approval from Project Sponsor that project meets SDG4 and SDG8
- The APIs discussed in 3.3 are implemented such that they seamlessly integrate into the web application
- Three pathways of financial knowledge as stated in 2.2.1 are implemented with full functionality as discussed in 2.1
- Gamification features as outlined in 2.1 and 2.2 are fully implemented, this includes all leaderboard, avatar and shop features discussed.
- Have a 70% test coverage as this falls inline with industry standards^[1]

To achieve the criteria within a timely manner, team members must communicate on a regular basis (convene at least once a week) to confirm progress made on each of the features.

4.2 Extended Features / Future Milestones

If development continues beyond the end of the project due date additional features should be included:

- Ability for users to set other users as friends to monitor each other's progress
- Deploy the web app online
- Tutorial when a user first creates an account to guide new users on how to navigate the app, and direct them to relevant learning materials
- Admin users have the ability to create new quizzes through an extension of Django's administrator interface

4.3 Security Concerns

To ensure proper security for users and prevent unauthorised access to their data, strong password policies must be enforced when creating an account as well as two-factor authentication being recommended. This is achieved using Google's sign-in API, as Google provides 2-step authentication for its account holders.^[2]

For future scope, security audits must be regularly undertaken to evaluate potential risks. This can be done with penetration testing.

5.0 Team Charter & Rules of Engagement (10%)

5.1 Rules of engagement

Each team member should use prior experience to participate equally in contributing to the concept, development and testing stages of the project, adhering to industry protocols and standards. Code should be accompanied with detailed documentation to allow each team member to work on any part if necessary as outlined in section 6.4.1. Consistent engagement is expected and can be shown through: regular attendance to meetings and meeting deadlines.

When taking a managerial role, the team member should compile with all expectations and responsibilities outlined as follows. The Project Manager (PM) will manage the scope and the overall timeline for the project, promote communication between team members and with external entities, and be the final decision maker within the team. This role will be occupied by Allia (week 1-6) and Grace (week 7-12). The Documentation Lead is responsible for managing the report writing so it meets the PM's deadlines, editing & submitting the reports and taking meeting minutes. This role will be occupied by Allia for the duration of the project. Finally the Technical Lead will manage coding so that it meets the PM's deadlines, assigning coding tasks, setting up the skeleton code, releasing/submitting the code and approving merge requests. This role will be occupied by Josh (weeks 1-7) and Alex (weeks 8-12).

Three times during the development process, there will be formal feedback opportunities via the Formative Feedback Form. During this time, individuals can evaluate their performance thus far and identify improvement areas in themselves and others to provide constructive criticism as a starting point for personal development.

5.2 Team Targets

To work effectively and cooperatively within the team structure as to create a web application solution to the given problem statement such that it achieves all evaluation criteria outlined in 4.1. This will be evaluated through the use of the Formative Feedback Forms and the 4.1 success criteria. By the end of the project, the team hopes to have had at least two face-to-face meetings with the Project Sponsor to discuss progress and future timelines as well as two instances of formative feedback from the Primary Stakeholder.

5.3 Individual Goals

5.2.1 Allia - By week 6, my goal is to become a proficient project manager, gauged through feedback forms collected from each team member.

5.2.2 Prayush - Within the completion of the coding phase, I aim to have elevated my front-end development skills through online resources and applying them throughout the coding phase.

5.2.3 Josh - By week 6, I aim to gain experience in setting up and managing Git repositories, which will be measured by successfully managing these Git repositories.

5.2.4 Alex - Throughout the coding phase, I aim to further my back-end development skills in Python's Django framework by successfully modelling and implementing features to be managed via a database, measured by the efficiency of the database system.

5.2.5 Bipin - By week 12 to have gained proficiency in using Git for effective code management and collaboration, focusing on version control. Measuring my proficiency by successfully aiding the management of code and collaboration with team members.

5.2.6 Grace - By the project's end, to have improved my API knowledge and gained experience in GitLab for CI/CD in development cycles. I plan to assess my progress by successfully integrating APIs and utilising GitLab for CI/CD.

5.4 Dispute/Member Disagreement Protocol

After attempting individual conflict resolution, the disagreement should be reported to the most relevant Lead who will step in as a mediator. If this fails, a team vote should be held, with the final escalation step being to brief the Project Sponsor on said disagreement for their input.

6.0 Project Management and Methodology (20%)

6.1 Change in Management Procedures

In the context of this project, a change in management would consist of any of the three Lead position titles: Documentation Lead, Technical Lead & Project Manager moving from one team member to another. This team member would take on all the responsibilities that that role requires as outlined in 5.1.

The team must be given a 1 week notice minimum of the change either verbally during a meeting or via the dedicated Teams Software Engineering Group 1 channel (Teams Channel). If conveyed verbally, this must be included as part of the meeting's notes document and posted to the Teams Channel. If the change is outlined in section 5.1, the notice period and its conditions can be ignored as all team members have had notice since the start of the project. All team members must know who the new Lead will be for the transition period to occur.

The transition period begins 1 week before the elected change-over date (when the old Lead steps down and the new one assumes the role). During the transition period, the old Lead will pass along all old documentation i.e future schedules & meeting plans as well as any current material being created in this period. Any new material created, should be a collaborative effort between the two individuals to allow for a smoother transition.

On the identified date, the new Lead will resume full control over that role's responsibilities.

6.2 Team Meetings & Record Keeping

Meetings are to be scheduled at the Project Manager's discretion via the WhatsApp chat or Teams Channel such that the maximum number of group members are available for the given circumstances.

Two days before the meeting, the Project Manager should post an Agenda to either the WhatsApp chat or/and Teams Channel outlining the key talking points.

The Documentation Lead will keep minutes during the meeting. If they are not present they, or the Project Manager, can assign a 'minutes man' in their stead. These minutes should be written up with all the necessary information including date, time & location, attendees, agenda, notes and action points. This document will be posted onto the WhatsApp chat within 48hrs and on the Teams Channel within 1 week after the meeting's date but only after the dispute window has closed. The document is to be posted on the WhatsApp chat first as it can be assumed that all team members will check the WhatsApp chat more regularly than the Teams Channel. In this instance, the Teams Channel is used as backup storage for the finalised version of these documents as well as updating our Project Sponsor.

All team members have 24hrs from the document being posted to the WhatsApp Chat to make any disputes about what has been written. This will follow the dispute process outlined 5.4.

This approach will be the baseline procedure for all documentation practices, unless otherwise stated in this document (PDD), with the Documentation Lead having authority to alter if needed with the Project Manager's approval.

6.3 Communication Between Stakeholders

6.3.1 Between Group and Project Sponsor

The Project Manager will be the primary point of contact between the team and the Project Sponsor, however any team member is able to contact them with queries or requests. The Microsoft Teams Group 1 channel should be used for any communications between the two entities, with email via the University of Surrey's Outlook being an inferior alternative when necessary. The Project Sponsor's email is as followed: s.kazamia@surrey.ac.uk

6.3.2 Between Group and Primary Stakeholder

The Project Sponsor will be the primary point of contact between the team and Dominic Brockes. This is to prevent other team members from contacting the Primary Stakeholder individually with general queries and prevent them from needing to repeat themselves in multiple answers. This ensures clarity and transparency with the communication remains high. The expected form of communication between the entities is via the Teams Group Channel; the team submits questions to the Project Sponsor and the Sponsor replies with the required answers from the Stakeholder.

6.3.3 Between Group and All Other Stakeholders

Currently no end-users are expected, therefore no communication channels are required at this time. Should this be expected to change this needs to be re-evaluated, The Project Manager will be the primary contact between the team and all other stakeholders via email. However for the companies that developed the APIs that we will be implementing, whilst the Project Manager will be the primary contact, the Technical Lead will provide input where necessary as a secondary contact.

6.4 Group Development

6.4.1 Documentation Practices

Code for the backend should be well documented. Documentation should include how the code works, what it is intended to do and how it can be interfaced with. Documentation should be written alongside the code to ensure that it remains up-to-date.

6.4.2 Code Style

PEP8^[2] style to be used, linter will be included in the pipeline. Spaces are the preferred indentation method in PEP8. Adhere to DRY throughout code. Comments should be provided to ensure it is clear what your code does

6.4.3 Source Version Control

For each task that needs to be completed, an issue will be created on the repository Gitlab page. A

separate branch and merge request will be made so that a team member(s) can work on the issue without working directly on the trunk branch. Once the code is complete, the technical lead should be notified so they can review. Anyone who has worked on a particular branch must approve the merge request, as well as the technical lead.

6.4.4 Quality Assurances

It will be the responsibility of the technical lead to review code pushed to the repository, ensuring the code follows the style guide, is commented appropriately and functions as intended. Tests should be written alongside the code to ensure functionality. The linter and any written tests should be executed as part of the pipeline, as well as any further checks needed in the future.

7.0 Work Plan (15%)

A visual representation of this work plan can be found under section 9.7 as Figure 1 and a full itinerary in Figure 2.

7.1 Research & Planning

During the research stage, the team is developing the Web Application's attributes at a conceptual level. This includes: design features, methodologies & frameworks that could be used, and possible APIs that could be integrated within the app. At the end of this stage, all individuals will have a strong understanding of the above aspects and be able to write their individual PDD section. This stage is expected to last 1 week, and be completed by Week 3.

The planning stage will consist of team members using the structure of the PDD and Project Pitch (PP) to help plan the specifics of the project. This includes the Project Proposal meeting with the Project Sponsor as well as the meeting with the primary stakeholder Dominic Marcus Financial Consulting to further help fine-tune the PDD and their ideas before coding begins. Team meetings will be vital in this stage to help keep high levels of communication between members. This stage should be completed by Week 7.

7.2 Handover of Roles

At Week 6 of the project there will be a team lead handover for the Project Manager. The role will be transferred from Allia to Grace to give those who volunteered an equal opportunity to be active in this role.

Another team lead handover will occur in Week 7 during the Development stage. The Technical Lead role will be transferred from Josh to Alex for identical reasoning for the handover of the Project Manager.

A staggered approach is taken to allow for a smoother transition for the wider team, as well as allow for the 1st Technical Lead to take a more dominant role within the team when development of the Web App begins. More details can be found in 5.1 for the roles' responsibilities, and 6.1 for specifics on handover protocols.

7.3 Development

This stage starts with a week of final research week to fill in any gaps that are identified during the last few stages. This also allows time for the Technical Lead to set-up the programming environment for all team members. This set-up step includes writing the skeleton code, allowing the wider team to hit the ground running when coding begins. The skeleton code should include: blank HTML pages, URL paths for said pages, blank testing files, and setting up the API support within the App's settings.py file.

Coding should begin on Week 7. The team will be divided into front and backend groups to start developing. Figure 3 in 9.7 will be used as the specific plan/flow chart that the team will follow and be assigned to with tasks such as quiz backend, quiz front end and database seeding. Whilst coding, brief Final Audit Report (FAR) notes should be taken by each member as well as test writing for the code that they write.

7.4 Testing & Deployment

This stage begins on Week 10 and should be completed by the beginning of Week 11. All outstanding features are either finished or removed from the product. The app will then be divided into sections and each team member will increase the test coverage for their assigned section with a target test coverage of 70% as stated and justified in section 4.1.

The Technical lead will deploy the finished product in Week 11. Details of the process can be found in section 3.4.1.

7.5 Final Audit Report Writing

The Documentation Lead would have started work on the FAR in Week 6 by outlining the basic structure it should follow. This is expanded on in Weeks 9-10 by the Documentation Lead as they begin writing the first draft, using the notes from the other team members to guide them.

In Week 11, all other team members will join the writing process as the Documentation Lead begins editing from oldest to newest written section. This is done to streamline the writing and editing process of the report.

7.6 Contingency Planning

Under this work plan, the PDD & PP will be completed 1 week ahead of schedule, thus creating a 1 week buffer zone to the first half of the project's timeline. Once coding begins, if any unforeseen circumstances arise, the Easter Break offers 3 weeks to bring the project back on schedule. In the worst case scenario, the development stage must end in Week 10 to allow for adequate time to complete the FAR document.

7.7 Task Assignment & Personal Goal Achievement

Prayush will be the key member designing the front-end of the web-app and thus, while still being full stack, will have more of an emphasis on the front-end to allow for his personal goal in 5.3 to be achieved. As Technical Lead, Josh will be able to complete his personal goal at the start of the development stage by managing git merge requests and assigning issues to group members in line with their personal goals. Bipin and Grace will work closely with both Technical Leads to learn more on utilising GitLab for CI/CD development cycles and code management without the need to take on a lead role. Alex and Grace will be assigned implementation of one API each, with Alex being assigned only back-end tasks such as creating the majority of the database backend to help accomplish his 5.3 goal. Other than these pre-established tasks, all other tasks will be assigned through volunteering or the relevant Lead assigning it to someone with consideration for the individuals preferences.

7.8 Non-Core Tasks

All non-core elements (those in 2.2.2) will be assigned once all core functionality has been established and if development has been completed within the timeframe given. However, if this project were to be worked on after the initial week 12 deadline, this too is when such tasks could be fully itemized and assigned.

8.0 Risk Analysis (10%)

In Risk Analysis, the objective is to identify, assess, and prioritise potential risks linked with the development of our web application. By comprehensively understanding these risks and implementing robust strategies to mitigate or manage them, we can significantly bolster the success and resilience of our project.

8.1 Risk Analysis Table

Below is an outline of the top 5 risks that could be identified at this time in rank order. Risks with the greatest potential to impede our progress and achievements are ranked higher. This was determined based on their Probability, which denotes their likelihood of occurrence, their Impact, which signifies the expected disruption if this were to happen, and their Severity, indicating to what

degree of potential harm could the risk cause. See Table 1 in 9.8 for a visual representation as well as a detailed analysis of all 12 identified risks.

8.1.1 Delay in Development Schedule

Prob: Low | Imp: High | Sev: High

Implement proactive project management techniques to monitor and track progress regularly. Break down project tasks into manageable segments and establish clear milestones and deadlines. Conduct regular progress reviews with the team to identify potential delays early on. Prioritise resources or adjust project timelines as needed to mitigate delays. Implement contingency plans to manage unforeseen challenges or setbacks effectively. Work collaboratively to find solutions that minimise the impact on project objectives and deadlines.

8.1.2 Ineffective communication

Prob: Low | Imp: High | Sev: High

To mitigate this risk, open communication will be encouraged, as well as the use of collaboration tools like Teams, Google Docs, and proactively resolve conflicts as they arise.

8.1.3 Difficulty Implementing APIs

Prob: Low | Imp: High | Sev: High

To mitigate this risk, allocated contingency time will be given for API integration, as well as exploring alternative APIs or having backup APIs, if necessary.

8.1.4 Scope Creep

Prob: Low | Imp: Medium | Sev: Medium

Define clear project scope and requirements from the outset and implement change control processes to manage any requested changes effectively. Regularly review project scope with stakeholders to ensure alignment and minimise scope creep.

8.1.5 Arguments and Disputes Among Team

Prob: Low | Imp: Medium | Sev: Medium

To mitigate this risk, open communication will be encouraged, as well as the use of collaboration tools like Teams, Google Docs to resolve conflicts proactively.

8.2 Lower Ranked Risks

The identified lower rank risks, while possessing potential impact, are considered low in probability and severity due to effective mitigation strategies and proactive measures already in place. Some risks, such as legal or intellectual property issues, may not be as pertinent given the nature of our academic endeavour. However, the team will continue to conduct necessary checks to ensure the avoidance of these risks. Given that our focus primarily revolves around verification processes, mitigating these risks becomes manageable through proper utilisation of the Agile methodology, GitLab as a reliable version control tool, and proper documentation practises.

9.0 Appendix

9.1 Project Charter

[1] A. Dwiastanti, "Financial literacy as the foundation for individual financial behavior.," *Journal of Education and Practice*, <https://eric.ed.gov/?id=EJ1083664> (accessed Feb. 23, 2024).

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Conclusion: Most young adults lack financial knowledge & having better knowledge has been linked to better financial outcomes

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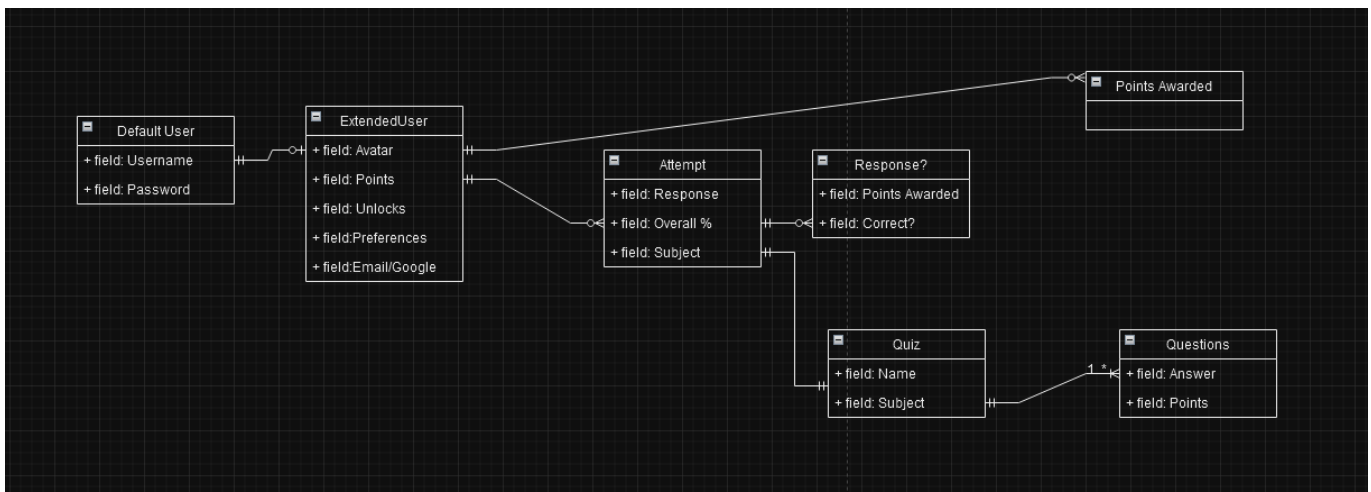
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9.3 Technical Approach

Figure 1



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9.4 Evaluation and Success Criteria

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Conclusion: Test coverage should be a minimum of 60-70%

C. Arguelles, M. Ivanković, and A. Bender, "Code coverage best practices," Google Testing Blog, <https://testing.googleblog.com/2020/08/code-coverage-best-practices.html> (accessed Feb. 17, 2024).

Conclusion: 60% is "acceptable", 75% is "commendable" and 90% is "exemplary"

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9.5 Team Charter

No external references or extra resources needed.

9.6 Project Management

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9.7 Work Plan

Figure 1

	1	2	3	4	5	6	7	8	9	10	11	12
Doc Lead				PDD Editing		FAR Struc			FAR Writ		FAR Edit	
Tech Lead						Set up					Depoly	
Team Mem1		research	PDD writ	Project Pitch		Soft Res	Coding			Finish app	FAR Writ	
Team Mem2		research	PDD writ	Project Pitch		Soft Res	Coding			Finish app	FAR Writ	
Team Mem3		research	PDD writ	Project Pitch		Soft Res	Coding			Finish app	FAR Writ	
Team Mem4		research	PDD writ	Project Pitch		Soft Res	Coding			Finish app	FAR Writ	
Team Mem5		research	PDD writ	Project Pitch		Soft Res	Coding			FAR Writ		
Team Mem6		research	PDD writ	Project Pitch		PP Editing	Coding			Finish app	FAR Writ	

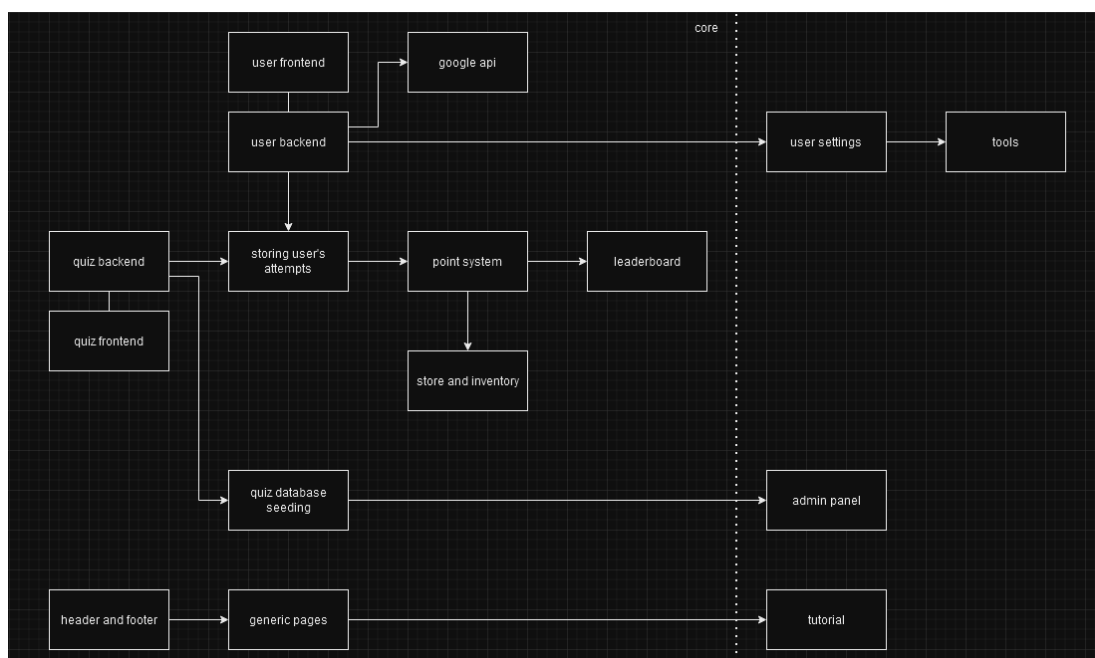
Everyone is a team member, but the 2 that are also leads must fulfil their specific shown tasks as well as their individual member one.

Figure 2

ITINERARY

1. Project Management
 - 1.1 Establish project concept & Team member's roles
 - 1.2 Research requirements
 - 1.2.1 Frameworks, APIs, Design
 - 1.3 Make Project Definition Document
 - 1.3.1 Assign each team member a section
 - 1.3.2 Write first draft
 - 1.3.3 Add any major alteration
 - 1.3.4 Formatting and minor edits
 - 1.4 Project Proposal
 - 1.4.1 Send over PDD (stage 1.3.4) to project sponsor
 - 1.5 Project Pitch
 - 1.5.1 Write and record pitch
 - 1.5.2 Send pitch for review
 - 1.5.3 Receive feedback
2. Software Development
 - 2.1 Final softwear researching (if required)
 - 2.2 Setting up version control
 - 2.3 Coding software
 - 2.3.1 Back & frontend team rotations established
 - 2.3.2 Coding the Web App
 - 2.3.2 Notes for Final Audit Report taken
 - 2.3.3 Unit Testing
 - 2.3.3.1 Record results & debug where necessary
 - 2.6 Deployment on Docker
3. Documentation
 - 3.1 Final Audit Report
 - 3.1.1 Section/Sub-section assignment
 - 3.1.2 Completing drafts (write, edit, review cycle)
 - 3.1.3 Submit
 - 3.2 Personal Review Report

Figure 3



9.8 Risk Analysis

Table 1

Event	Probability	Impact	Severity	Rank	Action
Delay in development schedule	Low	High	High	1	Implement proactive project management techniques to monitor and track progress regularly. Break down project tasks into manageable segments and establish clear milestones and deadlines. Conduct regular progress reviews with the team to identify potential delays early on. Prioritise resources or adjust project timelines as needed to mitigate delays. Implement contingency plans to manage unforeseen challenges or setbacks effectively. Work collaboratively to find solutions that minimise the impact on project objectives and deadlines.
Ineffective communication (between members)	Low	High	High	2	Encourage open communication, use collaboration tools like Teams, Google Docs, resolve conflicts proactively.
Difficulty in implementing APIs	Medium	Medium	Medium	3	Allocate contingency time for API integration, explore alternative APIs/ have backup APIs, if necessary.
Loss of code due to technical issues	Low	High	Medium	12	Maintain regular backups on GitLab and other remote drives, implement version control.
Team member leaving (unexpectedly)	Low	High	Medium	11	Cross-train team members, document work thoroughly, have contingency plans in place.
Inactive team members	Low	High	High	7	Encourage participation, redistribute workload, escalate to project sponsor if necessary.
Potential arguments and	Low	High	High	5	Encourage open communication, use collaboration tools like Teams, Google Docs, resolve conflicts proactively.

Disputes among team					
Scope Creep	Low	Medium	Medium	4	Define clear project scope and requirements from the outset and implement change control processes to manage any requested changes effectively. Regularly review project scope with stakeholders to ensure alignment and minimise scope creep.
Insufficient Testing	Low	High	High	8	Enforce rigorous testing protocols throughout the development lifecycle, encompassing unit, integration, and user acceptance testing. Allocate adequate time and resources for testing. Additionally, the designated Test Lead will oversee bug detection and resolution, ensuring the integrity of tests and eliminating redundancies.
Resource Constraints	Low	High	Medium	6	Monitor resource utilisation closely and allocate resources efficiently to avoid bottlenecks or burnout. Consider outsourcing certain tasks or reallocating resources as needed to maintain project momentum.
Legal or Intellectual Property Issues	Low	High	Medium	9	Conduct legal reviews to ensure compliance with intellectual property laws and regulations. Obtain necessary licences or permissions for third-party assets or intellectual property used in the application. Implement measures to protect the application's own intellectual property rights.
Changes in Stakeholder Requirements	Low	High	High	10	Regularly engage with the sponsor to seek clarification and guidance on the problem statement. Implement change management processes to evaluate any modifications or adjustments requested by the sponsor effectively. Prioritise sponsor feedback and incorporate changes in a manner that minimises disruption to the project timeline and deliverables, while ensuring that the project continues to meet the sponsor's expectations.

Going past the 10,000 server calls limit with the Mortgage Calculator discussed in 3.3.2	Low	Low	Low	13	The Technical Lead will monitor this free usage of the API such that the team doesn't reach the limit. Team members should take the utmost care to make as few server calls as possible during the development process.
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